

Jay Janaskar

Robot Learning & Visual Computing

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Profile

Computer science master's student at NTNU specialising in visual computing and robot learning, finishing June 2027 — master's thesis with SINTEF Ocean on tactile SLAM, and an exchange semester at ETH Zurich in 3D vision and digital humans. Works day to day in Python and PyTorch with Isaac Lab, Docker and Linux on GPU clusters. Looking for work in robot learning, 3D vision or machine learning, in Norway or abroad.

Education

Master's degree, Computer Science – NTNU Trondheim

Aug 2025 – Jun 2027 (expected)

Specialisation: Visual Computing.

- Master's thesis with SINTEF Ocean, Aug 2026 – Jun 2027 (in progress): "Touch to Map: Tactile SLAM for 3D Object Reconstruction". Combines GelSight tactile sensors with RGB-D point clouds, using active touch to cut the noise in the reconstructed models.

Master's exchange programme, Computer Science – ETH Zurich

Feb 2026 – Aug 2026

Exchange semester at D-INFK. Courses: 3D Vision, Digital Humans, Shape Modeling and Geometry Processing.

Bachelor's degree, Computer Engineering – OsloMet University

Aug 2022 – Jun 2025

- Bachelor's thesis with SINTEF Digital: built API endpoints with FastAPI and MongoDB to connect Koala UI, our team's interactive frontend, to Crocodile, a Python library for entity linking.

Exchange programme – Nelson Mandela University, South Africa

Jul 2024 – Nov 2024

Courses: Emerging Technologies, Network Optimisation and Security.

Vocational Certificate, ICT – Gand VGS, Sandnes

Aug 2018 – Jun 2020

Foundations in Linux, the command line and Python.

Projects

DribbleAMP: Adversarial Motion Priors for Football Dribbling

Spring 2026

Digital Humans, ETH Zurich – team of two

- Trained humanoid dribbling policies with Adversarial Motion Priors, using a learned discriminator instead of hand-written style rewards. Built the Unitree G1 environment and the Isaac Lab backend, and ran PPO on MimicKit with 1024 parallel environments in the Newton GPU simulator.
- Measured what the style reward buys and costs: AMP produced a natural, upright gait, but at an equal training budget the no-AMP baseline halved speed and angle error. Code at github.com/ErykHalicki/DribbleAMP [↗].

Extending Functional Scene Graphs to Include Articulated Object States

Spring 2026

3D Vision, ETH Zurich – team of three

- Added articulation state to functional 3D scene graphs, so a robot tracks a changing scene without rescanning. Grounding DINO, SAM and Qwen3-VL detect objects and parts, merged across RGB-D views.
- Fusing repeated interactions cut axis-angle error by 36% (4.99° to 3.19°) and axis-line position error by 19% (0.139 m to 0.112 m). Joint-type accuracy rose from 86.6% to 100%.
- Wrote the proposal and final presentations, reviewed the team's code, and documented the pipeline.

Experience

Apprentice – Sandnes Kommune

Aug 2020 – Jul 2022

- Configured and maintained networks, and installed 10+ switches for Sandnes Rådhus.
- Managed users, devices and access control in Active Directory, Azure AD, Airwatch, Google Admin and eFeide.

Intern – Tietoevry

2019 – 2020 (6 weeks)

- Built robotic process automation (RPA) with UiPath, using OCR to reduce manual processing.

Skills

- **Programming:** Python, C/C++, TypeScript/JavaScript, Java, C#
- **Robot Learning & ML:** PyTorch, reinforcement learning (PPO, Adversarial Motion Priors), Isaac Lab, MimicKit, Newton GPU simulator
- **3D & Vision:** RGB-D reconstruction, point clouds, 3D scene graphs, Grounding DINO, SAM, vision-language models
- **Web & Data:** React, Next.js, FastAPI, Spring Boot, PostgreSQL, MongoDB
- **Tools:** Docker, Linux, Git, SLURM (ETH Euler and NTNU Idun GPU clusters)

- **Languages:** Norwegian (native), English (bilingual), Marathi and Tulu (spoken at home), Hindi (conversational)

Volunteering

Developer – Make NTNU

Sep 2025 – Present

- Maintain the student organisation’s website.

Board member – Badminton OSI, the student sports club

Aug 2023 – Jan 2025

- Managed tournament registration for club members.